

Odd-Cycle-Free Latin Squares: Products and Small-Order Examples

2 October 2026

Compact research paper

Not externally peer reviewed

Abstract

A Latin square is FFF when every permutation induced by two distinct rows, columns, or symbols has only even cycles. This focused account proves that the three-bit odd-cycle witness pattern of a direct product is the componentwise Boolean OR of the factor patterns. A direct product is group-isotopic precisely when both factors are group-isotopic. Odd orders greater than one are excluded, while a group table is FFF exactly when the group is a 2-group. Pattern preservation through two-element congruence blocks supplies an explicit non-group-isotopic FFF loop of order 8 and, by products, an infinite family. Separately labelled finite computations record the small-order boundary and the order-8 census. An explicit order-12 table disproves this project's former power-of-two-only conjecture. The proofs, computational dependencies, and limits of these conclusions are kept distinct. The unrestricted order-18 existence problem remains open.

Keywords. Latin squares; quasigroups; direct products; cycle structure; reproducible computation.

2020 Mathematics Subject Classification. 05B15, 20N05.

Status, attribution, and evidence boundary

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This compact paper is one of three reading editions of the same research project, not a third independent publication. The *Selected Results* edition retains the former extended compact companion. The *Full Research Report* retains the comprehensive technical account. All three use the 30 September baseline, with a selected extension of 2 October 2026: an order-nine symmetry obstruction proved in *Selected Results* and the *Full Report*. The present selection is deliberately narrower: it develops the direct-product theorems, a binary-congruence construction, and small-order examples. It does not append every later construction or search campaign.

Statements with written proofs are distinguished from those labelled *Computational Theorem*. The latter depend on exact finite evidence; the order-8 census also depends on the completeness of the cited external enumeration. Fresh checks of displayed tables do not rerun an entire census. Internal proof review, independent implementations, and machine proof checking are not independent human refereeing.

The order-12 witness refutes this project's former power-of-two-only conjecture. Order 10 is excluded in the full report by a separate mathematical reduction and exact computation; that result is background here, not a proof supplied by this compact paper. None of the three editions resolves unrestricted order 18, proves a complete higher-order main-class census, or establishes literature priority. Changes in selection and presentation do not change the evidence status of the underlying results.

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1 Introduction

A Latin square has three interchangeable coordinate roles: row, column, and symbol. Comparing two lines in one role gives a permutation of a complementary coordinate set. The question considered here is whether any such permutation has a nontrivial odd cycle. Recording the answer in all three views gives an ordered pattern in $\{F, T\}^3$, where T denotes the presence of an odd-cycle witness. The designation FFF means that no view has such a witness. Testing only rows, or only rows and columns, does not establish this condition.

The central structural statements concern direct products. Their induced permutations factor coordinatewise, and product-cycle lengths are least common multiples of the factor lengths. This yields the exact rule

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K).$$

A separate closure criterion shows that the product is group-isotopic if and only if both factors are. Both assertions include converses, and neither relies on a finite census. In particular, an FFF factor preserves the pattern of its partner, and a nongroup factor cannot become group-isotopic by taking a direct product.

Two small examples make the scope concrete. A binary extension of the cyclic group of order 4 gives a nonassociative FFF loop of order 8. Its properties are proved, rather than inferred from an exhaustive scan. At order 12, an explicit table supplies a computationally checked FFF example. Consequently, existence is not confined to powers of two, even though every group-isotopic FFF square has power-of-two order.

For comparison, the finite data record contains no FFF square of order 6 and exactly 230 FFF main classes at order 8, of which 225 are not group-isotopic. The census is relative to the external classification and data of McKay, Meynert, and Myrvold [MMM07, McK]. These counts and the written proofs are different types of evidence. The product construction gives higher-order existence, not injectivity on main classes or a higher-order classification.

This paper is a focused reading edition of a larger research dossier. It does not claim priority for standard permutation arguments, group-isotopy criteria, or their formulations. The *Selected Results* edition adds further constructions and structural results; the *Full Research Report* supplies the detailed order-10 computational exclusion and precisely bounded order-18 search results. Those dependencies are not silently replaced by shorter arguments here.

2 Definitions and invariance

Let X be a set of size n . A *Latin square* on X is a map $L: X \times X \rightarrow X$ such that every row and every column is a bijection. For $a, c, u \in X$, define the row, column, and symbol-line bijections

$$r_a(c) = L(a, c), \quad q_c(a) = L(a, c), \quad s_u(c) = \text{the unique } a \text{ such that } L(a, c) = u.$$

For two lines in the corresponding view, put

$$\rho_{a,b} = r_b^{-1} r_a, \quad \kappa_{c,d} = q_d^{-1} q_c, \quad \sigma_{u,v} = s_v^{-1} s_u.$$

All three permutations act on an n -element complementary coordinate set.

Lemma 2.1. *If the two lines are distinct, the corresponding induced permutation is fixed-point-free.*

Proof. A fixed point of $\rho_{a,b}$ would give $L(a, c) = L(b, c)$ in one column, contrary to the Latin property. The column case is identical. A fixed point of $\sigma_{u,v}$ would say that the same cell contains the two distinct symbols u and v . $\square$

Definition 2.2. A view of L *fails* if one of its induced two-line permutations has an odd cycle of length greater than one. Write

$$\text{pat}(L) = (\text{row}(L), \text{col}(L), \text{sym}(L)) \in \{F, T\}^3,$$

where T means that the view fails. A square is *FFF* if $\text{pat}(L) = \text{FFF}$.

The terminology “fails” is chosen from the perspective of the local-trade program: a T view supplies an odd-cycle trade, while an F view does not.

An *isotopy* independently permutes rows, columns, and symbols. A *parastrophy* permutes the three coordinate roles. Their combined orbits are the *main classes* (also called species).

Proposition 2.3. *Under isotopy, each of the three view bits is invariant. Under parastrophy, the three bits are permuted. Consequently, being FFF and having at least one failing view are main-class invariants.*

Proof. Let an isotopy act by permutations α, β, γ of rows, columns, and symbols. The transformed row maps satisfy $r'_{\alpha(a)} = \gamma r_a \beta^{-1}$, and hence

$$(r'_{\alpha(b)})^{-1} r'_{\alpha(a)} = \beta(r_b^{-1} r_a) \beta^{-1}.$$

Thus row cycle lengths are unchanged. The column calculation conjugates by α , and the symbol calculation conjugates by β . Parastrophy only permutes which coordinate pair is called row, column, or symbol. $\square$

3 Elementary structural results

3.1 Odd orders and group tables

Proposition 3.1. *Every Latin square of odd order $n > 1$ has pattern TTT.*

Proof. By Lemma 2.1, every induced permutation between two distinct lines is fixed-point-free on an odd set. If all of its nontrivial cycles were even, their lengths would have even sum. Hence it has an odd cycle of length greater than one. This argument applies separately to any pair of distinct rows, columns, and symbols. $\square$

Let G be a finite group and let $L_G(x, y) = xy$ be its Cayley table.

Proposition 3.2. *The row permutation induced by rows a, b of L_G is left multiplication by $b^{-1}a$. The column permutation induced by columns c, d is right multiplication by cd^{-1} . The symbol permutation induced by symbols u, v is right multiplication by $u^{-1}v$. Each cycle length is therefore the order of the corresponding group element.*

Proof. The row identity is $r_b^{-1} r_a(x) = b^{-1} a x$. Likewise $q_d^{-1} q_c(x) = x c d^{-1}$. Finally, $s_u(c) = u c^{-1}$ and $s_v^{-1}(a) = a^{-1} v$, whence $s_v^{-1} s_u(c) = c u^{-1} v$. Left or right multiplication by g has orbits equal to cosets of $\langle g \rangle$, all of length $\text{ord}(g)$. $\square$

Corollary 3.3. *The Cayley table of a finite group G is FFF if and only if G is a 2-group.*

Proof. If G is a 2-group, every nonidentity element has even order, so Proposition 3.2 gives no nontrivial odd cycle. Conversely, if $|G|$ has an odd prime divisor, Cauchy’s theorem supplies a nonidentity element of odd order; the corresponding induced permutations give a failing view. $\square$

3.2 A closure criterion for group isotopy

Fix a base row b of a Latin square L and define

$$H_L(b) = \{r_b^{-1}r_x : x \in X\} \subseteq \text{Sym}(X).$$

Proposition 3.4 (Basis-family closure criterion). *A Latin square L is isotopic to a group table if and only if $H_L(b)$ is closed under composition. This condition is independent of the chosen base row.*

Proof. Suppose first that $H = H_L(b)$ is closed. It contains the identity, has $|X|$ distinct elements, and is a finite submonoid of a symmetric group; hence it is a group. For each $z \in X$, the values $(r_b^{-1}r_x)(z)$, as x varies, are all distinct, since a column of L contains every symbol once. Thus H acts regularly on X .

Fix $z_0 \in X$. Relabel rows by $x \mapsto h_x = r_b^{-1}r_x$. Use the regular action to identify every column z with the unique $h_z \in H$ satisfying $h_z(z_0) = z$, and identify every relabeled symbol $r_b^{-1}(u)$ in the same way. The entry in row h_x and column h_z is represented by the unique element sending z_0 to $h_x h_z(z_0)$, namely $h_x h_z$. The resulting isotope is therefore the Cayley table of H .

Conversely, for a group table the family at base row b is the full left regular representation $\{\lambda_{b^{-1}x} : x \in G\}$ and is closed. Isotopy conjugates the family and relabels its members, preserving closure. The same argument for any base row proves base independence. $\square$

The same criterion can equivalently be written in the column or symbol view, by parastrophy.

4 Direct products

Let L be a Latin square on X and K a Latin square on Y . Their direct product $P = L \times K$ is the Latin square on $X \times Y$ defined by

$$P((x, y), (x', y')) = (L(x, x'), K(y, y')).$$

Lemma 4.1 (Product-cycle formula). *Let π and τ be permutations. If x lies on a π -cycle of length m and y lies on a τ -cycle of length n , then (x, y) lies on a cycle of length $\text{lcm}(m, n)$ under $\pi \times \tau$.*

Proof. The pair returns after $t > 0$ steps exactly when both $m \mid t$ and $n \mid t$. The least such t is $\text{lcm}(m, n)$. $\square$

Theorem 4.2 (Pattern product theorem). *For arbitrary Latin squares L and K ,*

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K)$$

componentwise.

Proof. Rows of P are indexed by (a, α) , and their row maps are $r_{(a, \alpha)}^P = r_a^L \times r_\alpha^K$. Hence the induced permutation between (a, α) and (b, β) is

$$\rho_{(a, \alpha), (b, \beta)}^P = \rho_{a, b}^L \times \rho_{\alpha, \beta}^K.$$

If one coordinate of the two product rows is equal, the corresponding factor is the identity. The identical calculation for column maps gives $\kappa^P = \kappa^L \times \kappa^K$.

For the symbol view,

$$s_{(u,\eta)}^P(c, d) = (s_u^L(c), s_\eta^K(d)),$$

and therefore $\sigma^P = \sigma^L \times \sigma^K$. Thus the same argument applies in all three views.

If a factor has a nontrivial odd cycle in one view, keep the other factor line fixed. The identity contributes cycle length 1, so the product contains the same odd cycle. Conversely, suppose a product induced permutation has an odd cycle of length $\ell > 1$. By Lemma 4.1, $\ell = \text{lcm}(m, n)$ for factor cycle lengths m, n . Both are odd, and at least one is greater than one. That factor already has a nontrivial odd cycle in the same view. This proves the Boolean OR formula in each view. $\square$

Corollary 4.3. *$L \times K$ is FFF if and only if both L and K are FFF. In particular, product with a finite 2-group table preserves the full pattern of the other factor.*

Proof. The first assertion is the all- F case of Theorem 4.2. For the second, combine the same theorem with Corollary 3.3. $\square$

Theorem 4.4 (Group-isotopy product theorem). *$L \times K$ is group-isotopic if and only if both L and K are group-isotopic.*

Proof. At a product base row (b, c) , the basis family factors as

$$H_{L \times K}(b, c) = H_L(b) \times H_K(c).$$

If both factor families are closed, their Cartesian product is closed. Conversely, both factor families contain the identity. If the product family is closed, then $(h_1, \text{id})(h_2, \text{id})$ lies in it for all $h_1, h_2 \in H_L(b)$, proving closure of $H_L(b)$; using (id, k_i) proves closure of $H_K(c)$. Apply Proposition 3.4. $\square$

Corollary 4.5. *Let L be a non-group-isotopic FFF Latin square of order n . For every $m \geq 0$, the product $L \times C_2^m$ is a non-group-isotopic FFF Latin square of order $n2^m$.*

Proof. The table of C_2^m is FFF by Corollary 3.3; apply Corollary 4.3 and Theorem 4.4. $\square$

5 Binary congruences and a nongroup example

A Latin operation is a quasigroup operation: each equation $a * x = b$ or $y * a = b$ has a unique solution. The two solution operations are left and right division. A congruence is an equivalence relation preserved by multiplication and both divisions.

We next use congruences in the quasigroup algebra with multiplication and the two division operations. All congruence blocks have equal size and form a Latin quotient.

Theorem 5.1 (Congruence obstructions). *Let L be a finite Latin square.*

- (i) *If L has a congruence with odd block size greater than one, then $\text{pat}(L) = \text{TTT}$.*
- (ii) *If the congruence blocks have size two and the quotient is Q , then $\text{pat}(L) = \text{pat}(Q)$ componentwise.*

Proof. For (i), choose distinct rows x, x' in one congruence block A and any column block B . The two left translations map B bijectively onto the same product block. Thus $L_{x'}^{-1}L_x$ preserves B and is fixed-point-free there. Since $|B|$ is odd, its restriction has a nontrivial odd cycle. Applying the same argument to the two relevant parastrophes proves the column and symbol bits.

For (ii), label every two-element block by $(i, 0), (i, 1)$, where $i \in Q$. The unique form of a binary Latin square gives a function $f: Q^2 \rightarrow \mathbb{F}_2$ such that

$$(i, a) * (j, b) = (i * j, a + b + f(i, j)).$$

Rows in the same quotient block induce only transpositions inside the column fibres. For rows in different quotient blocks, the lift above a quotient cycle of length ℓ is either two cycles of length ℓ or one cycle of length 2ℓ , according to its binary voltage. Thus an ambient odd cycle can occur only above an odd quotient cycle.

Conversely, suppose quotient rows i, i' induce an odd cycle C of length $\ell > 1$. Above a quotient column $j \in C$, the lifted row permutation comparing (i, a) with (i', a') has the form

$$(j, b) \mapsto (\pi(j), b + a + a' + f(i, j) + f(i', \pi(j))).$$

Its voltage around C is $a + a' + S$, because ℓ is odd, where S is independent of a, a' . Choose $a + a' = S$. The voltage is then zero, so the lift of C consists of two cycles of the same odd length ℓ . Hence a quotient row witness lifts to an ambient row witness. This proves equality of the row bits. The same complete argument in the parastrophes proves equality of the column and symbol bits. $\square$

The same theorem also gives a census-free nongroup example.

Theorem 5.2 (Explicit nongroup FFF loop). *On $\mathbb{Z}_4 \times \mathbb{F}_2$, define*

$$(i, a) * (j, b) = (i + j, a + b + \mathbf{1}_{(i,j)=(1,1)}),$$

with coordinates taken modulo 4 and 2, respectively. This is an FFF loop of order 8 that is not group-isotopic.

Proof. The identity is $(0, 0)$, and the first-coordinate fibres form a congruence with quotient C_4 . The quotient is FFF by Corollary 3.3, so binary pattern equality makes the loop FFF. It is not associative because

$$((1, 0) * (1, 0)) * (2, 0) = (0, 1), \quad (1, 0) * ((1, 0) * (2, 0)) = (0, 0).$$

At the identity base row, the basis family in Proposition 3.4 is the family of left translations. Closure would force $L_x L_y = L_{x*y}$ and hence associativity. The displayed failure is therefore a closure failure, so the loop is not group-isotopic. $\square$

6 Computational results

The following statements are exhaustive finite computations. Their frozen inputs, programs, commands, outputs, and hashes are specified in Section 8.

Computational Theorem 6.1 (Small orders). *The unique reduced Latin square of order 2 is FFF, and all four reduced Latin squares of order 4 are FFF. None of the 9408 reduced Latin squares of order 6 is FFF. More precisely, their patterns are*

$$6600 \text{ TTT}, \quad 936 \text{ TFF}, \quad 936 \text{ FTF}, \quad 936 \text{ FFT}.$$

The exact-cover generator loads no cached square list. Every generated square was checked to be Latin and reduced, all 9408 table strings were distinct, and the generated set

agrees exactly with the ANU reduced-order-6 data file [McK]. Proposition 2.3 therefore extends nonexistence from reduced squares to all Latin squares of order 6.

Computational Theorem 6.2 (Order-8 pattern census). *Across the 283657 order-8 main-class representatives, the pattern distribution is*

<i>Pattern</i>	<i>Count</i>	<i>Pattern</i>	<i>Count</i>	<i>Pattern</i>	<i>Count</i>	<i>Pattern</i>	<i>Count</i>
FFF	230	FFT	81	FTF	40	FTT	315
TFF	133	TFT	209	TTF	232	TTT	282417

Among the 230 FFF main classes, exactly 5 are group-isotopic and 225 are not group-isotopic.

The ordered non-symmetric labels refer to the representatives in the cited ANU file. Parastrophy permutes the three bits, so labels such as FFT are not themselves main-class invariants. The FFF count, the existence of every pattern, and the group-isotopy split are unaffected.

The scan tests every unordered pair of lines in all three views. A separate closure-family computation reproduces the 5/225 split using Proposition 3.4. The frozen ANU input is an exact byte match to the file served by the cited URL on August 9, 2026.

Corollary 6.3. *For every $m \geq 3$, Latin squares of order 2^m realize all eight patterns. Every such order also contains a non-group-isotopic FFF Latin square.*

Proof. Computational Theorem 6.2 supplies one order-8 square of each pattern. Take direct products with the FFF group table of C_2^{m-3} and use Theorem 4.2. For the second assertion, start from the explicit loop in Theorem 5.2; non-group-isotopy is preserved by Theorem 4.4. $\square$

This is an existence result only. It neither counts higher-order main classes nor asserts that products of distinct order-8 classes remain distinct.

7 An explicit order-12 counterexample

The following statement is a direct finite verification, not an inference from an incomplete search or a solver timeout.

Computational Theorem 7.1 (An order-12 FFF square). *The reduced Latin square displayed in Table 1 is FFF. Thus this project’s earlier working conjecture that nontrivial FFF squares exist only at powers of two is false.*

The check verifies every row and column and constructs the relative permutations for all $3\binom{12}{2} = 198$ line pairs. It rejects fixed points and odd cycles explicitly. The exact table is supplied so that this positive existence statement can be rechecked without an external census. This does not decide order 18.

L_{12}	0	1	2	3	4	5	6	7	8	9	10	11
0	0	1	2	3	4	5	6	7	8	9	10	11
1	1	0	3	2	5	4	7	6	10	11	8	9
2	2	3	0	1	7	6	5	4	9	8	11	10
3	3	2	1	0	6	7	4	5	11	10	9	8
4	4	7	5	6	8	11	9	10	3	0	2	1
5	5	6	4	7	9	10	8	11	2	1	3	0
6	6	5	7	4	10	9	11	8	1	2	0	3
7	7	4	6	5	11	8	10	9	0	3	1	2
8	8	9	10	11	3	2	1	0	6	7	4	5
9	9	8	11	10	0	1	2	3	4	5	6	7
10	10	11	8	9	2	3	0	1	5	4	7	6
11	11	10	9	8	1	0	3	2	7	6	5	4

Table 1: An explicit reduced order-12 FFF witness. Every row and column is a permutation; all 198 induced line-pair permutations across the three views have only even cycles.

8 Reproducibility and evidence

The source package separates general proofs from finite controls. Every mathematical block reused in this edition is selected from the frozen 30 September source, with its source hash recorded by the edition builder. The longer editions reuse those sources rather than maintaining competing versions of a theorem. The editorial selection does not upgrade a claim.

Finite computational dependencies. The small-order enumeration is recorded in `repro_runs/2026-04-26_small_order_witness_reruns/`. Its generator constructs the reduced tables rather than loading a cached result, and checks Latinness, reduction, and uniqueness. The order-8 census is recorded in `repro_runs/2026-04-26_order8_core/`, with a separate group-isotopy reconstruction in `repro_runs/2026-04-26_order8_closure_reconstruction/`. The frozen compressed census input has SHA-256 `e5980ff5bc4ac46cbbefd1f1333258d0905724c42a976c4afe61ea501a4ccf87`. Matching this hash identifies the input; it does not independently prove that the external census is complete.

Displayed examples and product controls. The order-12 table is the frozen `data/order12_first.json` from the non-power-of-two witness intake, SHA-256 `8b71d6ecaa6ab86a806a0660529b404d8f3667db2b58b82bf5c72fed010fa416`. The edition check compares the displayed table with that input and traverses all 198 line-pair permutations. It also constructs the order-8 loop directly from its formula, checks its closure failure, tests products with C_2 and C_2^2 , and checks the eight frozen pattern controls. These are small checks of the selected content, not fresh exhaustive order-8 or order-10 computations. The general product results rest on the proof of Theorem 4.2 and the subsequent results, not on these tests.

Rebuilding this edition. From the research repository root, the command

```
python3 -B tools/build_paper_editions.py \
--output .audit/three-editions-check
```

builds all three editions using the fixed evidence date, with shell escape disabled. The output includes source-selection records, PDF hashes, page counts, and compiler diagnostics. The companion `scripts/check_compact_examples.py` runs the finite checks above. The

source package is `manuscript/candidates/2026-10-02_e9_symmetry_review/`. Its finite checker is retained in the three-edition predecessor.

The project is distributed for critical examination at <https://noetheon.github.io/fff-latin-squares/>. Repository checks establish specified file and computation properties, not peer review, novelty, or a complete order spectrum.

9 Scope and open questions

The product theorems give a complete answer to two construction questions: the witness pattern combines by Boolean OR, and group isotopy holds exactly when it holds in both factors. They do not classify all FFF squares. The explicit nongroup loop shows that group tables are not the only source, and the order-12 witness rules out a power-of-two-only existence spectrum.

The existence of all eight patterns in orders 2^m , $m \geq 3$, uses finite order-8 starting examples. The nongroup FFF family in the same orders instead follows from the explicit loop and written proofs. No statement here counts distinct higher-order main classes or assumes the product map is injective on them.

Unrestricted order 18 remains open. A failed construction, a timeout, or an exclusion for one parametrized family is not a nonexistence proof. The full report's separately evidenced order-10 exclusion is not repeated in this paper. Further families, represented-class lower bounds, and bounded order-18 exclusions belong in the longer editions, where their hypotheses and computational dependencies can be stated in full.

The next mathematical questions are which additional constructions yield FFF squares, how restrictive the three views are together, and which existence orders can be decided rigorously. Before any journal submission, independent expert review and literature comparison remain necessary to assess correctness, significance, and priority. This short selection is intended to make that review manageable, not to substitute for it.

References

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